JTMK Project Report:

Smart Plant Monitoring System

Project Title:

Smart Plant Monitoring and Watering System

Team Members:

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Introduction:

This project focuses on automating plant care through a monitoring system that checks soil moisture, temperature, and light levels. It ensures timely watering and informs users about plant health using real-time data.

Objectives:

1. To monitor soil moisture and ambient temperature using sensors.
2. To automate plant watering using a mini water pump.
3. To send plant health updates via LCD and buzzer notifications.
4. To help students learn sensor integration with Arduino.

Tools and Technologies Used:

• Hardware: Arduino Uno, Soil Moisture Sensor, DHT11, Light Sensor (LDR), Water Pump, LCD, Buzzer
• Software: Arduino IDE, Fritzing
• Programming Language: C/C++

Features:

• Real-time soil and weather condition monitoring
• Automatic watering system when moisture is low
• LCD display with environmental status
• Alert buzzer for critical values

Circuit Design Diagram:

(Insert image here of the full system circuit)

Implementation Process:

1. Planning: Determined sensor thresholds and watering logic.
2. Hardware Assembly: Assembled sensors, pump, and LCD.
3. Programming: Developed code to automate sensor responses.
4. Testing: Simulated dry/wet conditions and validated output.
5. Improvements: Calibrated sensors and adjusted pump flow.

Results:

• Soil moisture below 30% triggers watering
• Status updates every 10 seconds
• User-friendly output display

Challenges:

• Pump voltage mismatches
• False sensor readings in wet soil
• Ensuring safe waterproofing

Future Enhancements:

• Add mobile app integration
• Use solar power for sustainability
• Data logging for long-term analysis

Conclusion:

The project successfully provided an autonomous and educational tool for plant care, showing real potential for household and school use.

References:

1. Arduino Moisture Sensor Tutorial –
https://howtomechatronics.com
2. DHT11 Sensor Guide –
https://randomnerdtutorials.com
3. Water Pump Integration –
https://circuitdigest.com